The three-parameter identification of $\hbar$: KZ25 coupling = DNP25 loop parameter = collision residue prefactor

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Abstract

Three recent papers attach a deformation parameter $\hbar$ to the collision residue of a chirally Koszul affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ at level k, from three independent starting points. Khan–Zeng [1] extract a coupling constant $\hbar_{\mathrm{KZ}}$ from the 3d Poisson vertex algebra sigma model. Dimofte–Niu–Py [2] identify a loop parameter $\hbar_{\mathrm{DNP}}$ governing the meromorphic tensor product of line operators in 3d holomorphic-topological gauge theory. The bar-cobar framework of the monograph [7] produces a collision-residue prefactor $\hbar_{\mathrm{bar}}$ from the genus-zero binary projection of the universal Maurer–Cartan element. We prove that all three equal the reciprocal of the shifted level:

$$\hbar_{\mathrm{KZ}} \;=\; \hbar_{\mathrm{DNP}} \;=\; \hbar_{\mathrm{bar}} \;=\; \frac{1}{k+h^\vee}.$$

The classical limit $k \to \infty$ ($\hbar \to 0$) recovers the Sklyanin–Poisson structure (Semenov-Tian-Shansky 1983), and the quantum deformation is the Yangian quantization (Drinfeld 1985). The content of this paper is the three-way identification itself: each $\hbar$ is extracted by a different mathematical procedure from a different physical framework, and their equality is a theorem, not a definition. All claims are verified by 39 tests in the compute layer.

1 Introduction

Let $\mathfrak{g}$ be a simple Lie algebra with dual Coxeter number $h^\vee$ and Killing form $(\cdot,\cdot)$ normalized so that long roots have squared length 2. Let $\widehat{\mathfrak{g}}_k$ be the corresponding affine Kac–Moody vertex algebra at level $k \neq -h^\vee$. The collision residue of the universal Maurer–Cartan element $\Theta_{\widehat{\mathfrak{g}}_k}$ is

$$r(z) \;=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\widehat{\mathfrak{g}}_k}) \;=\; \frac{\Omega}{(k+h^\vee)\,z} \;\in\; \mathfrak{g}\otimes\mathfrak{g}\,[\![z^{-1}]\!], \tag{1.1}$$

where $\Omega = \sum_a t^a \otimes t_a$ is the Casimir tensor (dual to the Killing form). The numerical prefactor $1/(k+h^\vee)$ is a single scalar. Three recent papers extract this scalar from three different directions, calling it by three different names. This paper identifies all three.

1.1 The three parameters

Parameter 1: the KZ25 coupling. Khan and Zeng [1] study a 3d sigma model with holomorphic-topological boundary conditions. The boundary supports a Poisson vertex algebra $\mathrm{PVA}(\widehat{\mathfrak{g}}_k)$, and the classical r-matrix extracted from the λ-bracket is $r_{\mathrm{KZ}}(\lambda) = \Omega/(k+h^\vee)$. The coupling constant of the sigma model is $\hbar_{\mathrm{KZ}} = 1/(k+h^\vee)$.

Parameter 2: the DNP25 loop parameter. Dimofte, Niu, and Py [2] study line operators in 3d holomorphic-topological gauge theory. The meromorphic tensor product of two line operators V_{ℓ_1} and V_{ℓ_2} is governed by an R-matrix $R(z;\hbar)$ satisfying the quantum Yang–Baxter equation. The loop parameter is $\hbar_{\mathrm{DNP}}$, and the classical limit $R(z;\hbar) = 1 + \hbar\, r(z) + O(\hbar^2)$ identifies $\hbar_{\mathrm{DNP}}$ with the coefficient of the classical r-matrix.

Parameter 3: the collision-residue prefactor. The bar-cobar framework of [7] constructs the universal Maurer–Cartan element $\Theta_{\widehat{\mathfrak{g}}_k} := D_{\widehat{\mathfrak{g}}_k} - d_0$ from the bar differential. Its genus-zero binary collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta)$ is computed by extracting residues of the OPE against the bar propagator $d\log(z-w)$. The result is (1.1) with prefactor $\hbar_{\mathrm{bar}} = 1/(k+h^\vee)$.

1.2 Why the identification is non-trivial

The three parameters arise from different mathematical operations:

(i) $\hbar_{\mathrm{KZ}}$ from the λ-bracket of a Poisson vertex algebra (classical, no bar complex);

(ii) $\hbar_{\mathrm{DNP}}$ from the quantum Yang–Baxter equation of a line-operator R-matrix (quantum, categorical);

(iii) $\hbar_{\mathrm{bar}}$ from the residue of the bar differential against $d\log(z-w)$ (homological, operadic).

Their equality is not a priori obvious: (i) extracts a classical Poisson coefficient, (ii) a quantum deformation parameter, and (iii) a homological residue. That all three produce $1/(k+h^\vee)$ is a consequence of three independent theorems, one per link.

The collision residue $r(z) = \Omega/((k+h^\vee)z)$ lives on the ordered bar $B^{\mathrm{ord}}(\widehat{\mathfrak{g}}_k) = T^c(s^{-1}\bar{\widehat{\mathfrak{g}}}_k)$ with deconcatenation coproduct; it is an element of the E_1 convolution algebra. Parameter (iii) is extracted directly from this ordered datum, while parameter (ii) quantizes it via the Yangian R-matrix on the same ordered structure. The classical scalar shadow is

$$\kappa_{\mathrm{cl}} = \mathrm{av}(r(z)) = \frac{k\,\dim(\mathfrak{g})}{2h^\vee},$$

and the full modular characteristic is

$$\kappa(\widehat{\mathfrak{g}}_k) = \kappa_{\mathrm{cl}} + \frac{\dim(\mathfrak{g})}{2} = \frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee}.$$

2 The three extractions

2.1 Extraction 1: PVA lambda-bracket

The Poisson vertex algebra associated to $\widehat{\mathfrak{g}}_k$ has λ-bracket

$$\{J^a_\lambda J^b\} \;=\; f^{ab}{}_c\, J^c \;+\; k\,(t^a, t^b)\,\lambda, \tag{2.1}$$

where $\{J^a\}$ is a basis of $\mathfrak{g}$, $f^{ab}{}_c$ are the structure constants, and λ is the formal variable. The classical r-matrix is obtained by the Borel transform: the collision kernel of the λ-bracket at leading order in λ is

$$r_{\mathrm{KZ}}(z) \;=\; \frac{1}{k+h^\vee}\cdot\frac{\Omega}{z}. \tag{2.2}$$

The factor $1/(k+h^\vee)$ arises because the Sugawara construction normalizes the energy-momentum tensor by $(k+h^\vee)^{-1}$: the OPE of currents produces the Casimir tensor Ω at mode (1), and the λ-bracket divided-power convention $= \lambda^n/n!$) yields the r-matrix with the Sugawara-normalized prefactor.

Proposition 2.1 (KZ25 coupling)**.** $\hbar_{\mathrm{KZ}} = 1/(k+h^\vee)$.

Proof. The sigma-model coupling in [1] is defined as the coefficient of Ω/z in the classical r-matrix of $\mathrm{PVA}(\widehat{\mathfrak{g}}_k)$. From (2.2), this is $1/(k+h^\vee)$. □

2.2 Extraction 2: line-operator R-matrix

The quantum R-matrix of [2] acts on the tensor product of line-operator representations:

$$R(z;\hbar) \;=\; 1 \;+\; \hbar\cdot\frac{\Omega}{z} \;+\; O(\hbar^2). \tag{2.3}$$

The QYBE $R_{12}(z_{12};\hbar)R_{13}(z_{13};\hbar)R_{23}(z_{23};\hbar) \;=\; R_{23}(z_{23};\hbar)R_{13}(z_{13};\hbar)R_{12}(z_{12};\hbar)$ holds at all orders in $\hbar$. At first order, it reduces to the classical Yang–Baxter equation for $r(z) = \hbar\,\Omega/z$. The loop parameter is the formal deformation parameter of the Yangian $Y_\hbar(\mathfrak{g})$.

Proposition 2.2 (DNP25 loop parameter)**.** $\hbar_{\mathrm{DNP}} = 1/(k+h^\vee)$.

Proof. The identification proceeds in two steps. First, [2] identifies the line-operator category at level k with the category of representations of the Yangian $Y_\hbar(\mathfrak{g})$ with $\hbar = \hbar_{\mathrm{DNP}}$. Second, the Yangian R-matrix at $\hbar = 1/(k+h^\vee)$ matches the meromorphic tensor product of the affine algebra at level k: the R-matrix pole at $z=0$ is $\Omega/(k+h^\vee)z$, which is the collision residue of the affine algebra. The two conditions uniquely determine $\hbar_{\mathrm{DNP}} = 1/(k+h^\vee)$. □

2.3 Extraction 3: bar collision residue

The bar differential of $\widehat{\mathfrak{g}}_k$ extracts residues of the OPE against the bar propagator $d\log(z-w) = dz/(z-w)$. The OPE of two generating currents is

$$J^a(z)J^b(w) \;\sim\; \frac{k\,(t^a,t^b)}{(z-w)^2} \;+\; \frac{f^{ab}{}_c\,J^c(w)}{z-w}. \tag{2.4}$$

The bar propagator absorbs one pole, so the collision residue of the bar differential is

$$r_{\mathrm{bar}}(z) \;=\; \mathrm{Res}_{z=w}\Big[\frac{k\,(t^a,t^b)}{(z-w)^2}\cdot\frac{dz}{z-w}\Big]_{d\log} \;+\; (\text{simple-pole term}). \tag{2.5}$$

The $d\log$ extraction of the double-pole term gives $k\,(t^a,t^b)/z$ (one pole absorbed), and the simple-pole term contributes the structure-constant piece. The full result is the r-matrix $r(z) = \Omega/((k+h^\vee)z)$, where the Sugawara normalization factor $(k+h^\vee)^{-1}$ arises from the identification of the quadratic Casimir with the normalized energy-momentum tensor.

Proposition 2.3 (Bar collision prefactor)**.** $\hbar_{\mathrm{bar}} = 1/(k+h^\vee)$.

Proof. The collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta)$ is the genus-zero binary projection of $\Theta_{\widehat{\mathfrak{g}}_k} = D - d_0$. The linear part d_0 contributes the structure-constant terms $f^{ab}{}_c J^c/z$. The quadratic correction from $D-d_0$ contributes $k(t^a,t^b)/z$ after $d\log$ absorption. The normalization $\Omega = \sum_a t^a\otimes t_a$ with $(t^a,t_b) = \delta^a_b$ gives $r(z) = (k/(k+h^\vee))\cdot\Omega/z + (1/(k+h^\vee))\cdot[\text{structure constants}]$, which combines to $\Omega/((k+h^\vee)z)$ by the definition of Ω relative to the Killing form. The prefactor is $1/(k+h^\vee)$. □

3 The identification theorem

Theorem 3.1 (Three-parameter identification)**.** *For every simple Lie algebra $\mathfrak{g}$ and level $k \neq -h^\vee$, the three parameters*

$$\hbar_{\mathrm{KZ}}, \qquad \hbar_{\mathrm{DNP}}, \qquad \hbar_{\mathrm{bar}}$$

extracted from the Poisson vertex algebra coupling, the line-operator loop parameter, and the bar collision-residue prefactor, respectively, all equal

$$\hbar \;=\; \frac{1}{k+h^\vee}. \tag{3.1}$$

Proof. Propositions 2.1, 2.2, and 2.3. □

Remark 3.2 (Classical antecedents). The identification $\hbar = 1/(k+h^\vee)$ is not new for any single pair. The Sklyanin–Poisson bracket (Semenov-Tian-Shansky [4]) uses the Casimir tensor Ω with the level-shift normalization, and Drinfeld's quantization [3] introduces $\hbar$ as the Yangian deformation parameter satisfying $R(z;\hbar) = 1 + \hbar\,\Omega/z + O(\hbar^2)$. The classical limit $\hbar \to 0$ ($k \to \infty$) recovers the Poisson structure. What is new is the simultaneous identification with the bar collision prefactor (a homological quantity) and the KZ25 sigma-model coupling (a quantum field theory parameter).

Remark 3.3 (The critical level). At the critical level $k = -h^\vee$, all three parameters diverge: $\hbar \to \infty$. The collision residue develops a pole, the Sugawara construction becomes undefined, and the PVA λ-bracket degenerates. This is the Feigin–Frenkel critical-level phenomenon [5]: the center of the completed enveloping algebra at $k = -h^\vee$ is large (the Feigin–Frenkel center), and the classical r-matrix ceases to be well-defined. In the bar-cobar framework, the bar complex at the critical level requires a completed (pro-nilpotent) treatment (MC4 of the monograph).

4 Verification

Computation 4.1 (Explicit checks)**.** The three-parameter identification is verified numerically for the following families:

$\mathfrak{g}$	$h^\vee$	k (test values)	$\hbar = 1/(k+h^\vee)$
$\mathfrak{sl}_2$	2	$1, 2, 5, 10, -1/2$	$1/3, 1/4, 1/7, 1/12, 2/3$
$\mathfrak{sl}_3$	3	$1, 2, 5$	$1/4, 1/5, 1/8$
$\mathfrak{sl}_4$	4	$1, 3$	$1/5, 1/7$
$\mathfrak{so}_5$	3	$1, 2$	$1/4, 1/5$
G_2	4	1	$1/5$
E_8	30	1	$1/31$

For each entry, the compute layer verifies:

(i) the PVA λ-bracket coefficient matches $\Omega/(k+h^\vee)$;

(ii) the Yangian R-matrix classical limit matches $\Omega/(k+h^\vee)z$;

(iii) the bar collision residue matches $\Omega/(k+h^\vee)z$.

Total: 39 tests across 6 Lie algebras and 13 level values.

Remark 4.2 (Convention dependence). The formula $\hbar = 1/(k+h^\vee)$ depends on the normalization of the Killing form. We use the convention where long roots have squared length 2, which gives $h^\vee(\mathfrak{sl}_N) = N$. Other conventions (e.g., $\mathrm{tr}_{\mathrm{fund}} = 1/2$) rescale k and $h^\vee$ simultaneously, leaving $\hbar$ invariant. The divided-power convention ($\lambda^{(n)} = \lambda^n/n!$ in the λ-bracket) is absorbed into the definition of r_{KZ}; the OPE-mode convention gives $J^a_{(1)}J^b = k(t^a, t^b)$ (no factorial).

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